Summary in Graph

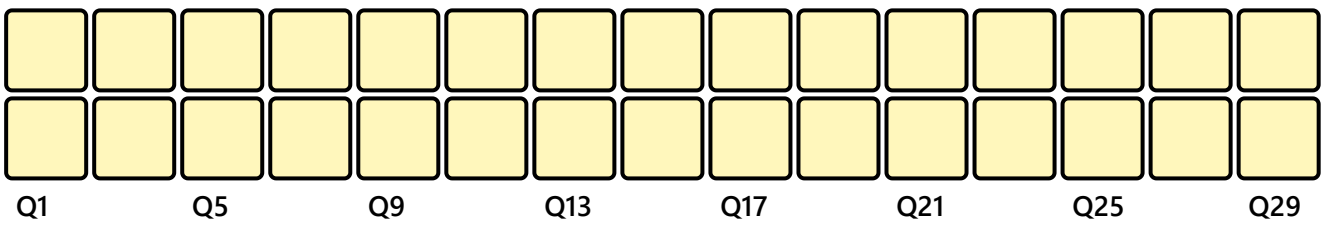
Exam Summary (GATE Overflow CS Test Series 2026 | Computer Networks | Test 1)

Qs. Attempted:	0 (0 + 0)	Correct Marks:	0 (0 + 0)
Correct Attempts:	0 (0 + 0)	Penalty Marks:	0 (0 + 0)
Incorrect Attempts:	0 (0 + 0)	Resultant Marks:	0 (0 + 0)

Total Questions:	30 (10 + 20)
Total Marks:	50 (10 + 40)
Exam Duration:	90 Minutes
Time Taken:	0 Minutes

Individual Question Statistics

Select Metric: Marks Obtained



* Average Stats based on first-attempt stats of Top 10% (7) out of 65 users

EXAM RESPONSE EXAM STATS FEEDBACK

Technical

Q #1 Multiple Choice Type Award: 1 Penalty: 0.33 Computer Networks

Which of the following link protocols best describes classical Ethernet?

A. Frequency-Division Multiple Access (FDMA)
B. Time-Division Multiple Access (TDMA)
C. Token-passing
D. Carrier Sense Multiple Access (CSMA)

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #2 | Multiple Select Type | Award: 1 | Penalty: 0 | Computer Networks

Which of the following is/are the services offered by UDP? (Mark all the appropriate choices)

A. Reliability
B. Multicast
C. Connection-less service
D. Congestion control

Your Answer: | Correct Answer: B;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #3 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Computer Networks

A computer is connected to the network of Computer Science Department which is part of the University network. The University network in turn is part of the internet. The largest network in which the IP address of the computer must be unique is:

A. the subnet to which the computer belongs
B. the Computer Science Department network
C. the University network
D. the Internet

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #4 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Computer Networks

A router has four half duplex Ethernet interfaces each operating at 600 Mbps. Ethernet frames are at least 84 bytes long. The maximum packet processing time at the router for wire-speed forwarding to be possible is (in microseconds)?

A. 0.56
B. 2.8
C. 0.14
D. 0.28

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #5 | Numerical Type | Award: 1 | Penalty: 0 | Computer Networks

An IP datagram of size 1200 bytes arrives at a router. The router has to forward this packet on a link whose MTU (maximum transmission unit) is 120 bytes. Assume that the size of the IP header is 20 bytes. The number of fragments that the IP datagram will be divided into for transmission is _____

Your Answer: | Correct Answer: 13 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #6 | Numerical Type | Award: 1 | Penalty: 0 | Computer Networks

Assume that source S and destination D are connected through three intermediate routers. If a is the number of times each packet has to visit the data link and b is the number of times each packet has to visit the network layer during a transmission from S to D, $2a + 3b =$ _____

Your Answer: | Correct Answer: 31 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #7 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Computer Networks

Which of the following mapping is CORRECT with respect to $IPv4$?

	Destination Address		Type
I	01:00:5E:32:10:2C	a	Broadcast Address
II	2C:22:15:13:54:C2	b	Multicast Address
III	FF:FF:FF:FF:FF:FF	c	Unicast Address

A. I – b, II – c, III – a
B. I – c, II – b, III – a
C. I – a, II – c, III – b
D. None of the above

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #8 | Multiple Select Type | Award: 1 | Penalty: 0 | Computer Networks

When a UDP segment arrives at a host, in order to direct the segment to the appropriate socket, the operating system's network stack uses which of the following fields? (Mark all the appropriate choices)

A. the source IP address
B. the destination IP address
C. the source port number
D. the destination port number

Your Answer: | Correct Answer: B;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #9 | Multiple Select Type | Award: 1 | Penalty: 0 | Computer Networks

When a TCP segment belonging to an existing connection arrives at a host, in order to direct the segment to the appropriate socket, the operating system's network stack uses which of the following fields? (Mark all the appropriate choices)

A. the source IP address
B. the destination IP address
C. the source port number
D. the destination port number

Your Answer: | Correct Answer: A;B;C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #10 | Multiple Select Type | Award: 1 | Penalty: 0 | Computer Networks

Which of the following is/are NOT correct? (Mark all the appropriate choices)

A. Full-duplex ethernet use CSMA/CD for collision detection
B. A dedicated switch port is required for each full-duplex node
C. Hubs can only run in half duplex mode
D. Half-duplex Ethernet shares a collision domain and provides a lower effective throughput than full-duplex Ethernet

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #11 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Computer Networks

A network with CSMA/CD protocol in the MAC layer is running at 3 Gbps over a 2 km cable with no repeaters. The signal speed in the cable is 2×10^8 m/sec. The minimum frame size for this network should be?

A. 30000 bytes
B. 7500 bytes
C. 15000 bytes
D. 60000 bytes

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #12 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Computer Networks

A and B are the only two stations on an Ethernet. Each has a steady queue of frames to send. A and B attempt to transmit a frame, collide, and B wins the first backoff race. At the end of this successful transmission by B, both A and B attempt to transmit and collide. The probability that A wins the second backoff race is:

A. $5/8$
B. $1/16$
C. $1/8$
D. $3/16$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 09sec | Discuss

Q #13 | Numerical Type | Award: 2 | Penalty: 0 | Computer Networks

Consider that 24 machines need to be connected in a LAN using 5-port Ethernet switches. Assume that these switches do not have any separate uplink ports. The minimum number of switches needed is _____

Your Answer: | Correct Answer: 8 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #14 | Numerical Type | Award: 2 | Penalty: 0 | Computer Networks

Consider a simplified time slotted MAC protocol, where each host always has data to send and transmits with probability $p = 0.3$ in every slot. There is no backoff and one frame can be transmitted in one slot. If more than one host transmits in the same slot, then the transmissions are unsuccessful due to collision. The maximum number of hosts this protocol can support if each host has to be provided a minimum throughput of 0.21 frames per slot is _____

Your Answer: | Correct Answer: 2 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #15 | Numerical Type | Award: 2 | Penalty: 0 | Computer Networks

A pure ALOHA network transmits 100-bit frames on a shared channel of 100 Kbps. The number of frames that will probably survive if system produces 250 frames per second is _____ (rounded to the nearest integer)

Your Answer: | Correct Answer: 37:38 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #16 | Numerical Type | Award: 2 | Penalty: 0 | Computer Networks

A slotted ALOHA network transmits 250-bit frames using a shared channel with a 250 kbps bandwidth. The throughput (percentage-wise) if the system (all stations together) produces 200 frames per second(round off to 1 decimal digit) will be _____

Your Answer: | Correct Answer: 16.3:16.4 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #17 | Numerical Type | Award: 2 | Penalty: 0 | Computer Networks

A link has transmission speed of 5×10^5 bits/sec. It uses data packets of size 2000 bytes each. Assume that the acknowledgment has negligible transmission delay and that the processing delays at nodes are negligible. The efficiency of the stop-and-wait protocol in this setup is exactly 35%. The value of the one way propagation delay (in milliseconds) is _____

Your Answer: | Correct Answer: 29:30 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #18 | Numerical Type | Award: 2 | Penalty: 0 | Computer Networks

Consider two hosts A and B, connected by a single direct link of rate 2×10^7 bits/sec. The distance between the two hosts is 12000 km and the propagation speed along the link is 2×10^8 m/sec. Host A sends a file of $60,000$ bytes as one large message to host B continuously. Let the transmission and propagation delays be x milliseconds and y milliseconds respectively. Then the value of $2x + 3y =$_____

Your Answer: | Correct Answer: 228 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #19 | Numerical Type | Award: 2 | Penalty: 0 | Computer Networks

A 300 Mbps link is carrying traffic from a number of sources. Each of them generates an on-off traffic stream; when the source is on, the rate of traffic is 12 Mbps, and when the source is off, the traffic rate is zero. The duty cycle, which is the ratio of on-time to off-time, is $1:4$. When there is no buffer at the link, the minimum number of sources that can be multiplexed on the link so that link capacity is not wasted and no data loss occurs is X. Assuming that all sources are synchronized and the link is provided with a large buffer, the maximum number of sources that can be multiplexed so that no data loss occurs is Y. The value of $X + Y =$ _____

Your Answer: | Correct Answer: 150 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #20 Numerical Type | Award: 2 | Penalty: 0 | Computer Networks

Host P is sending data to host Q over a full duplex link. P and Q are using the sliding window protocol for flow control. The send and receive window sizes are 4 packets each. Data packets $(\text{sent only from } P \text{ to } Q)$ are all 800 bytes long and the link capacity is 160 Mbps. Acknowledgment packets $(\text{sent only from } Y \text{ to } X)$ are very small and require negligible transmission time. The propagation delay over the link is $120\ \mu s$. The maximum achievable throughput (in Mbps rounded to one decimal points) in this communication is _____

Your Answer: | Correct Answer: 91:92 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #21 Numerical Type | Award: 2 | Penalty: 0 | Computer Networks

Station A needs to send a message consisting of 12 packets to station B using a sliding window $(\text{window size } 4)$ and GBN error control strategy. All packets are ready and immediately available for transmission. If every 6th packet that A transmits gets lost (but no acknowledgments from B ever get lost), then the total number of packets that A will transmit for sending the message to B will be _____

Your Answer: | Correct Answer: 23 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #22 Numerical Type | Award: 2 | Penalty: 0 | Computer Networks

A 2 Mbps satellite link connects two ground stations. The altitude of the satellite is $32,604$ km and speed of the signal is 3×10^8 m/s. Assuming that the acknowledgment packets are negligible in size and that there are no errors during communication, the minimum packet size (in bytes rounded to one decimal point) required for a channel utilization of 35% for a satellite link using GoBack-63 sliding window protocol is _____

Your Answer: | Correct Answer: 603:608 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #23 Numerical Type | Award: 2 | Penalty: 0 | Computer Networks

A channel has a bit rate of 6 kbps and one-way propagation delay of 30 ms. The channel uses stop and wait protocol. The transmission time of the acknowledgment frame is negligible. To get a channel efficiency of at least 75%, the minimum frame size in bytes should be _____

Your Answer: | Correct Answer: 135 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #24 Numerical Type | Award: 2 | Penalty: 0 | Computer Networks

Host X sends a UDP datagram containing 5516 bytes of user data to host Y over an Ethernet LAN. Ethernet frames may carry data upto 1400 bytes. Size of UDP header is 8 bytes and size of IP header is 20 bytes. There is no option field in IP header. If the total number of IP fragments transmitted is a and the contents of offset field in the last fragment is b, $a + b =$_____

Your Answer: | Correct Answer: 693 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #25 Numerical Type | Award: 2 | Penalty: 0 | Computer Networks

In the $IPv4$ addressing format, if the number of networks allowed under Class A, $\text{Class B and Class C}$ respectively are 2^a, 2^b and 2^c, the value of $2a + 3b + 5c$ = _____

Your Answer: | Correct Answer: 161 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #26 | Numerical Type | Award: 2 | Penalty: 0 | Computer Networks

Every host in an $IPv4$ network has a 1 millisecond resolution real-time clock with battery backup. Each host needs to generate up to 4096 unique identifiers per second. Assume that each host has a globally unique $IPv4$ address. If you are to design a 60-bit globally unique ID for this purpose, the time period (in seconds) after which the identifiers generated by a host wrap around will be _____

Your Answer: | Correct Answer: 64:66 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #27 | Numerical Type | Award: 2 | Penalty: 0 | Computer Networks

A TCP message consisting of 2400 bytes is passed to IP for delivery across two networks. The first network can carry a maximum payload of 1000 bytes per frame and the second network can carry a maximum payload of 5000 bytes per frame, excluding network overhead. The IP overhead per packet is 20 bytes. The total IP overhead in the second network for this transmission will be _____

Your Answer: | Correct Answer: 60 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #28 | Multiple Select Type | Award: 2 | Penalty: 0 | Computer Networks

The subnet mask for a particular network is $255.255.0.0$. Which of the following pairs of IP addresses could belong to this network? (Mark all the appropriate choices)

A. $158.59.195.66$ and $158.59.131.33$
B. $16.39.22.2$ and $16.39.48.9$
C. $191.28.106.43$ and $191.29.202.54$
D. $35.8.129.55$ and $35.9.129.59$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #29 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Computer Networks

The routing table of a router is shown below:

Destination	Subnet Mask	Interface
142.72.43.0	255.255.255.128	Eth1
142.72.43.0	255.255.255.0	Eth2
190.32.21.1	255.255.255.254	Eth3
Default		Eth4

On which interface will the router forward packets addressed to destinations $142.72.43.16$ and $192.32.21.3$ respectively?

A. Eth2 and Eth3

B. Eth1 and Eth3
C. Eth1 and Eth4
D. Eth2 and Eth4

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #30 | Multiple Select Type | Award: 2 | Penalty: 0 | Computer Networks

Which of the following is/are true about Ethernet networks? (Mark all the appropriate choices)

A. Ethernet frames are usually larger than the minimum required size of 64 bytes to ensure good utilization of the network.
B. A bridge will forward a broadcast but a switch will not.
C. Hosts on different segments of a switched Ethernet network will only see packets to hosts on their segments or to the broadcast address.
D. When an Ethernet sender detects that the media is idle, it sends a jam signal onto the media to tell other devices not to transmit, and then it sends its packet.

Your Answer: | Correct Answer: A;C | Not Attempted | Time taken: 00min 00sec | Discuss

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